

Winning Strategies (magnamagna)

Giorgio and William enjoy playing together the *Magnamagna* game, a typical Italian two-player game inspired by the well-known practice.

After collecting bribes from contestants², they first arrange them into a long line of N items of nominal value V_i for $i = 0 \dots N - 1$ (from left to right). Then, they play in turns each time choosing between two possibilities:

- take the leftmost item and gain its value ('L'),
- take the rightmost item and gain its value ('R').



Figure 1: The long line of bribes, to be taken in turns.

In any case, taking items from the interior of the line is never allowed.

Thanks to months of exhaustive research, Giorgio claims to have finally found the optimal strategy for this game, allowing him to collect the most possible money on every play. He thus decided to let William play first so as to compensate the strategic gap. Giorgio's strategy is indeed very simple: each time he chooses the move with the highest *smartness*. The smartness of a move is the quantity:

$$x_0 - \frac{x_1}{2} + \frac{x_2}{4} - \frac{x_3}{8} + \dots$$

where $x_0, x_1, \dots$ are the values in the line of items, taken **from the chosen end** on.

For example, consider the following line of items:

6 3 0 8

In this case, the smartness of move 'L' is $6 - 3/2 + 0/4 - 8/8 = 3.5$ and the smartness of move 'R' is $8 - 0/2 + 3/4 - 6/8 = 8$. Thus in this case, Giorgio would choose move 'R'. In the very unlikely case of the two moves having exactly the same smartness, Giorgio will let William choose for him.

 Real numbers closer than 10^{-6} are considered equal for the purpose of this task (we recommend to use double precision floating point numbers).

Despite Giorgio's claims, William is not that convinced of the optimality of this strategy. Help William collect the highest possible value, knowing that Giorgio always follows the above strategy!

 Among the attachments of this task you may find a template file `magnamagna.*` with a sample incomplete implementation.

Input

The first line contains the only integer N . The second line contains N integers V_i .

²Any reference to people and facts is purely fictional.



Output

You need to write a single line with a sequence of N characters 'L' or 'R', representing the moves taken by William and Giorgio.

Constraints

- $1 \leq N \leq 1000$.
- $0 \leq V_i \leq 100\,000$ for each $i = 0 \dots N - 1$.

Scoring

Your program will be tested against several test cases grouped in subtasks. The score in each subtask will be calculated as the **minimum** score obtained in any of its test cases, multiplied by the value of the subtask. The score in a test case will be **0** if Giorgio's moves in the produced sequence do not follow the strategy described above. Otherwise, it will be calculated as:

$$\max\left(\frac{W - W_{\text{avg}}}{W_{\text{opt}} - W_{\text{avg}}}, 0\right)$$

where W is the value collected by William through your program, W_{opt} is the maximum value that William can possibly collect, and W_{avg} is the value collected by William in average.³

- **Subtask 1 [5 points]**: Examples.
- **Subtask 2 [35 points]**: $N \leq 10$.
- **Subtask 3 [40 points]**: $N \leq 100$.
- **Subtask 4 [20 points]**: No additional limitations.

Examples

input.txt	output.txt
4 6 3 0 8	RLLR
5 84 75 42 25 51	LLRLR

Explanation

In the **first sample case**, William gains $8 + 3 = 11$ and Giorgio gains $6 + 0 = 6$.

In the **second sample case**, William gains $84 + 51 + 25 = 160$ and Giorgio gains $75 + 42 = 117$.

³The average is computed assuming that William chooses 'L' with probability 50% and 'R' also with probability 50%.